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A novel vegetation-water resistant soil moisture index for remotely assessing soil surface moisture content under the low-moderate wheat cover

Jibo Yue [a], Ting Li [a], Yang Liu [c,d], Jia Tian [e,*], Qingjiu Tian [f], Suju Li [b,*], Haikuan Feng [d], Wei Guo [a], Hao Yang [d], Guijun Yang [d], Hongbo Qiao [a,*]

[a] College of Information and Management Science, Henan Agricultural University, Zhengzhou 450002, China
[b] Key Laboratory of Emergency Satellite Engineering and Application , Ministry of Emergency Management, Beijing 100124, China
[c] Key Lab of Smart Agriculture System, Ministry of Education, China Agricultural University, Beijing 100083, China
[d] Key Laboratory of Quantitative Remote Sensing in Agriculture of Ministry of Agriculture China, Beijing Research Center for Information Technology in Agriculture, Beijing 100097, China
[e] School of Instrumentation and Optoelectronic Engineering, Beihang University, Beijing 100191, China
[f] International Institute for Earth System Science, Nanjing University, Nanjing 210023, China

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ABSTRACT

Soil surface moisture content (SMC) is one of the most critical and fundamental indicators of crop drought because water is vital for vegetation growth. In recent decades, satellite high-spatial-resolution optical multi-spectral remote-sensing sensors have entered a new stage. However, current optical remote sensing SMC indices (SMIs) are sensitive to cropland SMC and vegetation water content (VWC). From dry to saturated, the soil spectra decreased quickly as the SMC increased. For the vegetation spectra, the short-wave infrared (SWIR) bands decreased as the VWC increased. The differences in the responses of near-infrared (NIR) and SWIR bands to water may help distinguish soil moisture from mixed soil-vegetation spectra. This study focuses on (a) analyzing how soil and vegetation reflectance are affected by VWC and SMC, (b) designing a new vegetation-water resistant soil moisture index (VRSMI) based on the differences in the responses of NIR bands and SWIR bands to water, and (c) evaluating the potential of using VRSMI to remotely estimate SMC in low-moderate crop-covered regions. VRSMI uses SWIR1-SWIR2 as the numerator to detect the water absorption features and NIR$\times$(SWIR1)$^{0.5}$ as the denominator to weaken the vegetation effect. VRSMI and soil-VRSMI (VRSMIs) were tested using laboratory- and field-based spectra datasets. Our study presents the following conclusions: (1) The differences in the responses of NIR and SWIR bands to water help distinguish the effects of soil moisture on mixed soil-vegetation spectra. (2) VRSMIs can provide high-performance SMC estimates under low-moderate vegetation cover (NDVI < 0.55). (3) VRSMIs can be used for cropland SMC evaluation without considering crop VWC (laboratory-based spectra dataset: RMSE $= 0.07$–0.11, $R^2 = 0.96$–0.98; field-based spectra dataset: RMSE $= 0.17$–0.18, $R^2 = 0.63$–0.79). Our study indicates that VRSMI and VRSMIs can provide high-performance relative SMC estimates under low-to-moderate vegetation cover.

1. Introduction

Soil surface moisture content (SMC), or soil surface water content, is one of the most critical and fundamental indicators of crop drought because it is vital for vegetation growth (Amani et al., 2016; Ambrosone et al., 2020; Fan et al., 2022; Judge et al., 2021; Vo Quang et al., 2022; Zappa et al., 2022). In addition, SMC indicates groundwater conditions and links the energy and water exchange between the atmosphere and land surface. Thus, it plays a vital role in various hydrological processes (Lievens et al., 2015; Liu and Yang, 2022; Sahaar et al., 2022). As SMC

* Corresponding authors.
E-mail addresses: yuejb.ha@gmail.com, yuejibo@henau.edu.cn (J. Yue), liting@stu.henau.edu.cn (T. Li), liuyanghe810@cau.edu.cn (Y. Liu), tianjia@buaa.edu.cn (J. Tian), tianqj@nju.edu.cn (Q. Tian), lisujuff@vip.sina.cn (S. Li), fenghaikuan@nercita.org.cn (H. Feng), guowei@henau.edu.cn (W. Guo), yangh@nercita.org.cn (H. Yang), guijun.yang@163.com (G. Yang), qiaohb@henau.edu.cn (H. Qiao).

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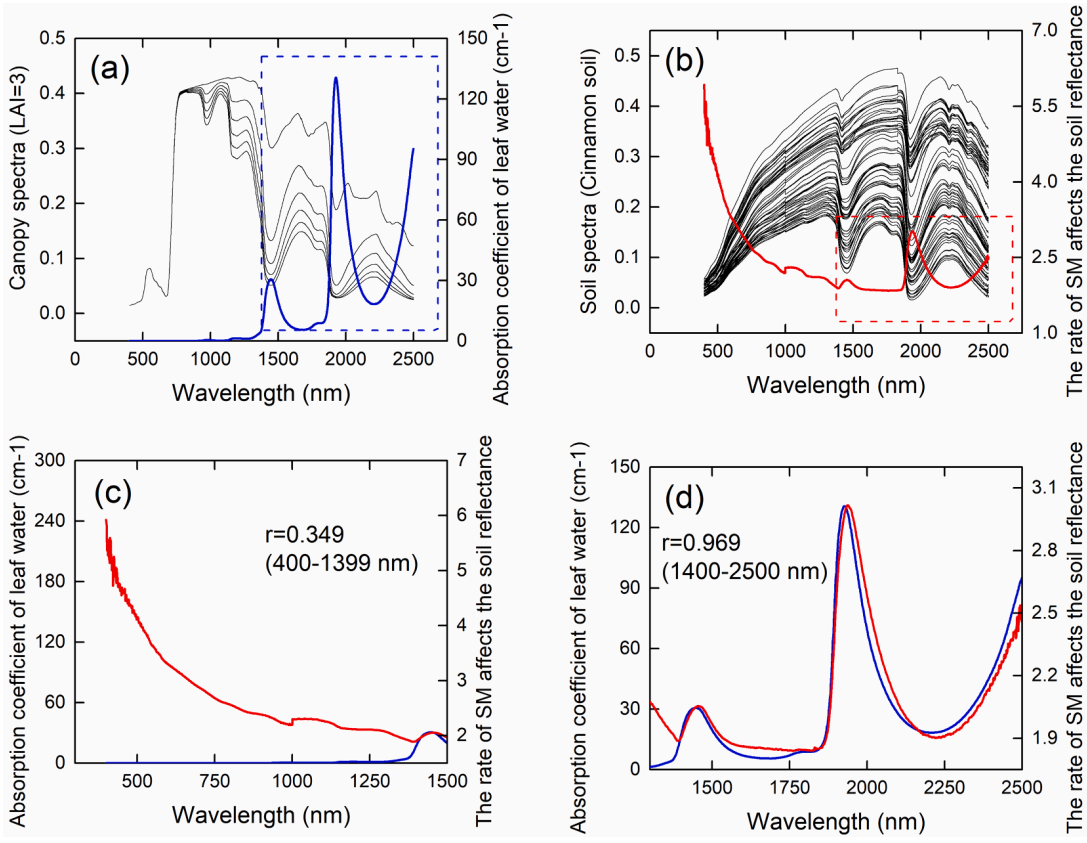


Fig. 1. Differences in the responses of NIR bands and SWIR bands to water. (a) Simulated vegetation canopy spectra and absorption coefficient of leaf water. (b) Soil spectra and the rate of SMC affects the soil reflectance. The absorption coefficient of leaf water and the rate of the SMC affects the soil reflectance at 400–1500 nm (c) and 1400–2500 nm (d). Note: Parameters of vegetation spectra: Cab = 40 μg/cm^2; Cw = 0.001:0.01:0.051 cm; Cm = 0.008 g/cm^2; LAI = 3 m^2/m^2; other parameters are listed in Table 1. The rate of SMC affects the soil reflectance was calculated based on the Lobell model (Lobell and Asner, 2002) (Eq (1)) and the Lab-1 dataset (cinnamon soil). The rate of SMC affects the soil reflectance is determined by SMC and soil reflectance values using the MATLAB 2021a Nonlinear Regression Tool (MathWorks, Natick, Massachusetts, USA; Levenberg-Marquardt method). The absorption coefficient of leaf water was obtained from the Spec_P5B data of PROSAIL.

plays a crucial role in agricultural irrigation management, water resource management, climate change, and environmental monitoring, its accurate and timely estimation has been studied intensively for a long time (Ihuoma et al., 2021; Lei et al., 2022; Vo Quang et al., 2022; Zhang et al., 2022).

Traditional manual methods for SMC measurement are not only time- and labor-consuming, but also unable to provide cropland SMC spatial distribution information (Susha Lekshmi et al., 2014). Remote sensing sensors from various platforms can capture the spectrum of the crop canopy, thereby providing information about the soil and vegetation (Akash et al., 2024; Liu et al., 2024b, 2024c, 2024a, 2023; Tian et al., 2021a; Yue et al., 2019; Zhu et al., 2024). Satellite remote sensing has recently gained popularity in cropland SMC mapping (Lievens et al., 2015; Peng et al., 2021; Wigneron et al., 2017; Zappa et al., 2022). Remote sensing-based cropland SMC estimation studies can be classified into four categories (Akash et al., 2024; Ali et al., 2015; Khandan et al., 2022; Peng et al., 2021): (a) optical remote sensing methods, (b) active and passive microwave remote sensing methods, (c) thermal infrared remote sensing methods, (d) synergistic methods, and (e) data assimilation methods.

Many optical remote sensing methods have been developed based on the SMC absorption features of the soil spectra (Bablet et al., 2018; Philpot, 2010; Sadeghi et al., 2017, 2015). Soil reflectance decreases as SMC increases because of water absorption (Lekner and Dorf, 2009; Philpot, 2010). Lobell and Asner (2002) developed a physical model to analyze the response of soil reflectance to the SMC.

$$R_i = R_{i,sat} + (R_{i,dry} - R_{i,sat}) \times \exp(-c \times \mathrm{SMC}) \quad (1)$$

where R_i is the soil reflectance for band i; $R_{i,dry}$ and $R_{i,sat}$ represent the dry and saturated soil spectral reflectance for band i, respectively; and c is the rate of SMC affecting the soil reflectance. However, physical models require prior knowledge that is often not readily available (e.g., soil roughness and texture) (Leone et al., 2019). Optical remote sensing SMC Indices (SMI) have been proposed to estimate SMC. SMI-based bare soil SMC estimation and mapping have been verified in many studies (Hardisky et al., 1983; Hunt and Rock, 1989; Jiaxin et al., 2024; Luo et al., 2023; Sadeghi et al., 2017; Yue et al., 2019).

Passive microwave remote sensing from satellites supports SMC mapping and improves crop drought modeling (Liang et al., 2021; Suman et al., 2021; Wigneron et al., 2017; L. Zhu et al., 2023; W. Zhu et al., 2023). However, the coarse spatial resolution limits agricultural irrigation management practices. Active microwave remote sensing (synthetic aperture radar (SAR)) can capture SMC at high spatial resolutions (Li and Wang, 2018; Moradizadeh and Srivastava, 2021; Santoro et al., 2021; Yadav et al., 2020). SAR remote sensing is sensitive to many soil parameters such as soil roughness, terrain, slope, and aspect (Li and Wang, 2018; Santoro et al., 2021; van Hateren et al., 2023; Yadav et al., 2020; Zhu et al., 2024). Despite this, many SMC products have been produced using microwave remote sensing techniques, such as the Soil Moisture Active and Passive missions (Reichle et al., 2019).

The basic concept of the thermal infrared remote sensing method is that the surface temperature is sensitive to SMC (Dall'Olmo and Karnieli, 2002; Karnieli et al., 2010). Using land-surface temperature (LST) to estimate SMC has been widely accepted (Chen and Liu, 2020; Karnieli et al., 2010; Rhee et al., 2010; Sun and Kafatos, 2007). As LST depends on many ambient atmospheric parameters, only LST is not suitable for directly mapping high-spatial-resolution SMC (Amazirh et al., 2019; Liu

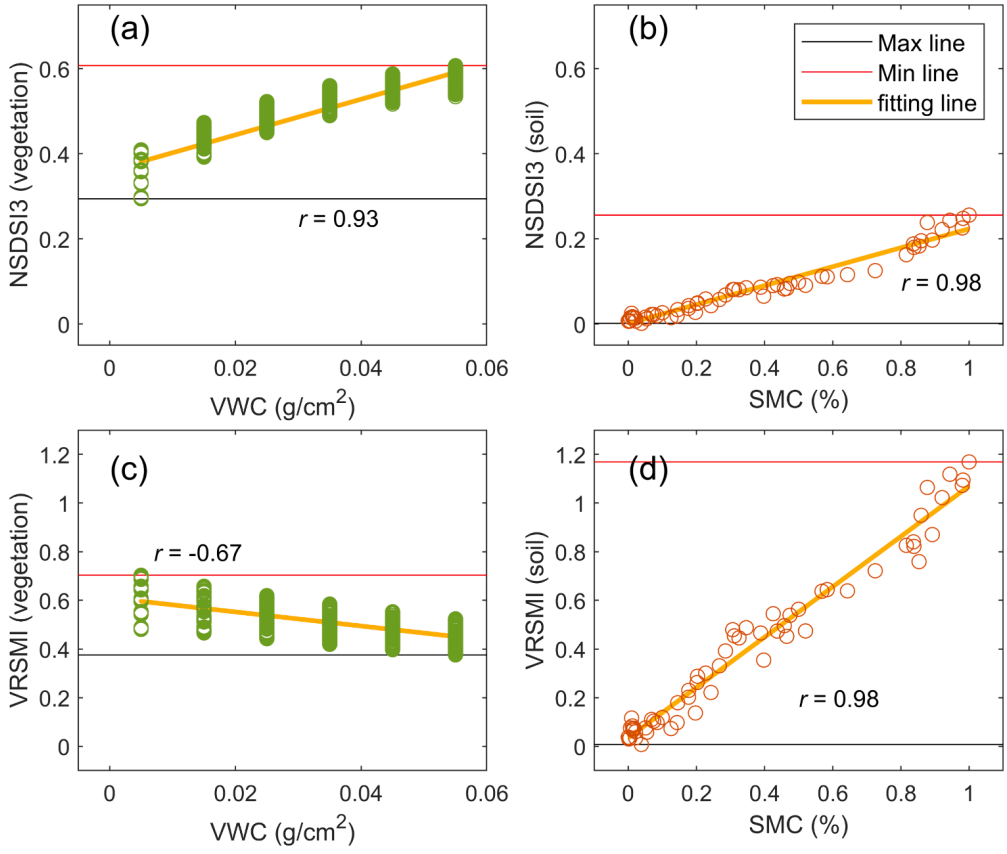


Fig. 2. Vegetation VRSMI is almost insensitive to VWC. Response of NSDSI3 (a)-(b) and VRSMI (c)–(d) to VWC and SMC. Note: LAIs = 5 m^2/m^2, Cm = 0.006 g/cm^2; for other parameters, see Table 1. NSDSI3=(SWIR1-SWIR2)/(SWIR1 + SWIR2) (Tian et al., 2021b; Yue et al., 2019).

et al., 2019). Synergistic methods use multi-remote sensing techniques to map SMC. Examples are LST and STR, LST and NDVI, passive microwave and SAR, optical remote sensing and SAR, or microwave and thermal infrared (Amazirh et al., 2019; Rahimzadeh-Bajgiran et al., 2013).

The data assimilation method estimates SMC using remote sensing observation data and machine learning, land surface, and global hydrological models (Li and Wang, 2018; Santoro et al., 2021; Wang et al., 2022; Yadav et al., 2020). Remote sensing observation data served as the input or intermediate variables of these models to obtain synthesized estimates of SMC (Lei et al., 2020; Tangdamrongsub et al., 2020; Tao et al., 2023; Xu et al., 2020). Many SMC products have been calculated based on data assimilation methods, such as SMAP level 4 (Reichle et al., 2019) and SMOS level 4 (Hoareau et al., 2014; Li et al., 2024).

In recent decades, satellite high-spatial-resolution multi-spectral remote-sensing sensors have entered a new stage. First, the ground spatial resolution of sensors is continuously improving; freely and widely used earth observation has a ground spatial resolution of 30 m (e.g., Landsat8 OLI, 2013, Landsat9 OLI-2, 2021) to 16 m (e.g., Gaofen1 WFV, 2013, Gaogen6 WFV, 2018) (Ya et al., 2024) to 10 m (e.g., Sentinel-2A MSI, 2015, Sentinel-2B MSI, 2017). Second, with two or more satellites operating, a 2-day (Gaofen1 + Gaofen6), 5-day (Sentinel-2A + 2B MSI) (ESA, 2015; Luo et al., 2023), and 8-day (Landsat8 + Landsat9) revisit time can ideally be achieved in most regions. Thus, optical remote sensing has the potential to provide cropland SMC information with high spatial and temporal resolution.

The reflected Earth radiation contains the soil and crop canopy water content information of a cropland field. The cropland SMIs of crop-soil mixed regions resulted from cropland SMC, vegetation water content (VWC), and fractional vegetation cover (FVC). Cropland VWC and FVC constantly change with many factors, such as crop growth stages, SMC, and farmland management. During the early growth stages of winter wheat (seeding to overwintering), the canopy height was less than 10 cm, and FVC ranged from 0 % to 60 % (Yang et al., 2023). An SMI that can weaken the wheat canopy moisture effect helps evaluate the cropland SMC and anomalous moisture events.

Few studies have evaluated the possibility of building an optical SMI that is insensitive to the VWC. From dry to saturated, the soil spectra decreased quickly as the SMC increased (Tian et al., 2021b; Yue et al., 2019). For the vegetation spectra, the short-wave infrared (SWIR) bands decreased as the VWC increased. The differences in the responses of near-infrared (NIR) and SWIR bands to water may help distinguish soil moisture from mixed soil-vegetation spectra. This study focuses on (a) analyzing how soil and vegetation reflectance are affected by VWC and SMC, (b) designing a new vegetation-water resistant soil moisture index (VRSMI) based on the differences in the responses of NIR bands and SWIR bands to water, and (c) evaluating the potential of using VRSMI to remotely estimate SMC in low-moderate crop-covered regions.

We propose a VRSMI and a soil-VRSMI (VRSMIs) for remote assessment of SMC in low-moderate FVC regions. VRSMI and VRSMIs were tested using three datasets: (i) Lab-1: two widely used leaf spectral datasets (lopex1993 and angers2003) and a soil spectra dataset; (ii) Canopy-2: a radiative transfer model (RTM)-based vegetation canopy spectra dataset from the widely used PROSAIL 5B RTM (https://teledetection.ipgp.jussieu.fr/prosail/) and a soil spectra dataset; and (iii) Field-3: field "wheat-soil" hyperspectral reflectance data and soil moisture measurements. Our study indicates that VRSMI and VRSMIs can provide high-performance relative SMC estimates under low-moderate vegetation cover.

Table 1
Parameters of the PROSAIL.

Parameter	Full name	Value *
N	Leaf structure index	1.8
Cw (cm)	Equivalent water thickness	0.005:0.01:0.055
Car (μg/cm^2)	Carotenoid content	8
Cbrown	Brown pigments	0
Cab (μg/cm^2)	Chlorophyll a + b content	30:10:70
Cm (g/cm^2)	Dry matter content	0.002:0.002:0.010
LAI	Leaf area index	0.0001, 0.1, 0.2, 0.4, 0.5, 0.8, 1.0, 2, 3, 5
hspot	Hot spot	0.1
ALIA (°)	Average leafinclination angle	30
tts (°)	Solar zenith angle	30
tto (°)	Observer zenith angle	10
psi (°)	Azimuth angle	90
psoil	Soil coefficient factor	real soil spectra: cinnamon soil: 60, brown soil: 59, alluvial soil: 59.

Note: * 0.005:0.01:0.055 represents the minimum, interval, and maximum Cw values, respectively. The soil spectra were obtained using laboratory-based soil measurements [Lab-1, soil spectra].

2. VRSMI and VRSMIs

2.1. Water absorption effect on soil and vegetation hyperspectral reflectance spectra

Fig. 1a shows that the absorption coefficient of leaf water and the vegetation spectral reflectance decreased with an increase in VWC (using Cw). Fig. 1b shows that the rate of SMC affects the soil spectral reflectance (Lobell and Asner, 2002), which decreases with an increase in SMC. The absorption features near 1.45 μm, 1.9 μm and 2.7 μm dominate the vegetation (Fig. 1a, blue dashed box) and soil (Fig. 1b, red dashed box) spectra.

The responses of the soil and vegetation SWIR bands to SMC and VWC were similar. The correlation coefficient between the leaf water absorption coefficients and the rate at which SMC affects the reflectance for 1400–2500 nm was 0.969 (Fig. 1d). Thus, the current SWIR-band-based SMIs are sensitive to SMC and VWC (such as NSDSI3=(SWIR1-SWIR2)/(SWIR1 + SWIR2)). The soil NIR spectra notably decreased with SMC (Fig. 1b); in contrast, the vegetation NIR spectra slightly decreased with increasing VWC (Fig. 1a). The correlation coefficient between the leaf water absorption coefficients and the rate at which SMC affects the reflectance for 400–1399 nm is 0.349 (Fig. 1c).

2.2. Proposed VRSMI based on the responses of NIR and SWIR bands to water absorption

We designed a VRSMI based on the differences in the responses of the NIR and SWIR bands to water (see Fig. 2). VRSMI uses SWIR1-SWIR2 as the numerator to detect the water absorption features and NIR×

Table 2
Statistical analysis of measured soil relative SMC.

Date	Numbers	Minimum	Maximum	Averaged
March 2, 2024	40	0.02	1	0.40
March 7, 2024	80	0.00	1	0.34
March 14, 2024	80	0.06	1	0.35
Total	200	0.00	1	0.35

Table 3
Current broadband SMIs and VWC indices.

Index	Equation	Reference
NDII1	(NIR − SWIR1)/(NIR + SWIR1)	(Hardisky et al., 1983)
NDII2	(NIR − SWIR2)/(NIR + SWIR2)	(Hardisky et al., 1983)
MSI1	SWIR1/NIR	(Hunt and Rock, 1989)
MSI2	SWIR2/NIR	(Hunt and Rock, 1989)
STR(SWIR2)	(1 − SWIR2)^2/(2 × SWIR2)	(Sadeghi et al., 2017)
NSDSI1	(SWIR1 − SWIR2)/SWIR1	(Yue et al., 2019)
NSDSI2	(SWIR1 − SWIR2)/SWIR2	(Yue et al., 2019)
NSDSI3	(SWIR1 − SWIR2)/(SWIR1 + SWIR2)	(Yue et al., 2019)

Note: NIR, SWIR1, and SWIR2 represent the NIR (0.83 μm), short-wave infrared bands at 1.6 μm, and 2.2 μm.

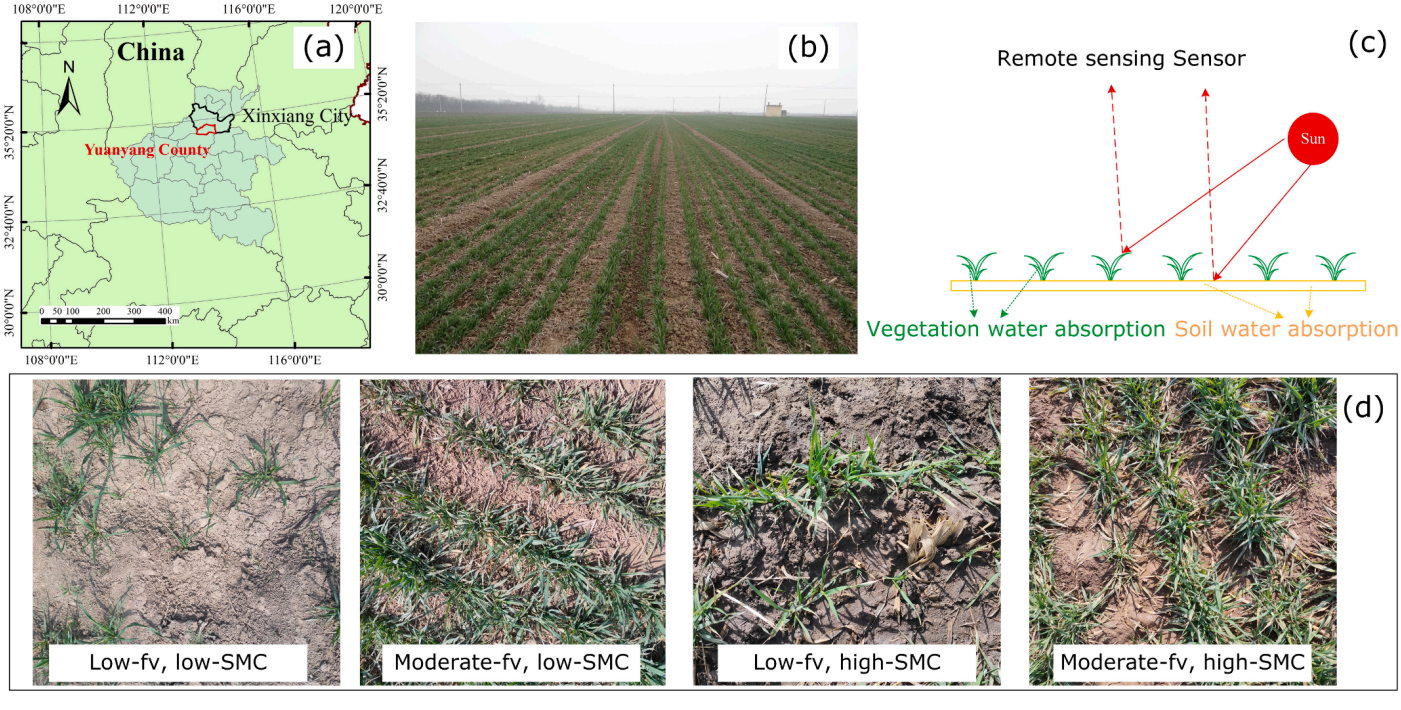


Fig. 3. (a) Map shows the Yuanyang County of Xinxiang City, China. (b) Cropland mixed landscape with winter-wheat FVC = about 20 %. (c) Diagram of a remote sensing mixed cropland pixel. (d) Field "wheat-soil" spectra and SMC collection scenes.

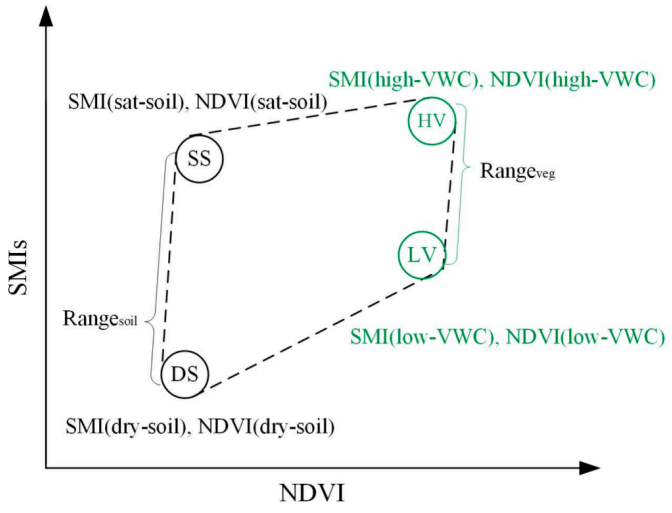


Fig. 4. Calculation methods of RVS.

Table 4
Numbers of samples for SMC model calibration and validation (cinnamon soil).

Dataset	NDVI	Scenes	Number	
			calibration	validation
Lab-1	–	bare soil	–	–
	>0.4	Plant leaf	–	–
Canopy-2	<0.25	bare soil	8, 869	5, 923
	0.25–0.4	low FVC	10, 264	6, 807
	0.4–0.55	moderate FVC	9, 868	6, 602
	<=0.55	low-moderate FVC	29, 001	19, 332
Field-3	<0.25	bare soil	–	22
	0.25–0.4	low FVC	–	44
	0.4–0.55	moderate FVC	–	64
	<0.55	low-moderate FVC	–	130
	>=0.55	high FVC	–	70

Note: All Field-3 samples were used for SMC model validation. The estimated SMCs greater than 1 (below 0) were replaced by 1 (0).

$(SWIR1)^{0.5}$ as the denominator to weaken the vegetation effect.

$$VRSMI = \frac{SWIR1 - SWIR2}{NIR \times (SWIR1)^{0.5}} \quad (2)$$

where NIR, SWIR1, and SWIR2 represent the NIR (0.83 μm) and SWIR bands at 1.6 μm and 2.2 μm, respectively. Fig. 2 shows the response of NSDSI3 = (SWIR1-SWIR2)/(SWIR1 + SWIR2) (Tian et al., 2021b; Yue et al., 2019) and VRSMI to the increase in SMC and VWC. Vegetation VRSMI was almost insensitive to VWC because vegetation SWIR1-SWIR2 and $NIR \times (SWIR1)^{0.5}$ decreased with VWC. VRSMI (dry soil) = 0 (Fig. 2d) because SWIR1-SWIR2 was much less than $NIR \times (SWIR1)^{0.5}$ (Fig. 2a, d). Soil VRSMI increases with SMC because (a) SWIR1-SWIR2 increases with SMC, and (b) $NIR \times (SWIR1)^{0.5}$ decreases with SMC. The ranges of NSDSI3 for vegetation and soil were 0.27–0.58 and 0.01–0.26, respectively. This indicates that NSDSI3 is also sensitive to VWC (Fig. 2a). In contrast, the VRSMI (0.01–1.17) increased with the SMC and was almost insensitive to the VWC (Fig. 2c, d, approximately 0.43–0.61).

2.3. VRSMIs

During the early growth stages of winter wheat (seeding to overwintering), the canopy height was less than 10 cm, and FVC ranged from 0 % to 60 %. As the VRSMI is almost insensitive to VWC (Fig. 2c, d), vegetation effects can be removed based on the linear spectral unmixing analysis (LSMA) theory:

$$VRSMI_0 = VRSMIv \times FVC + VRSMIs(1 - FVC) \quad (3)$$

where VRSMIv = 0.56, and VRSMIs represent vegetation- and soil-VRSMI. $VRSMI_0$ represents the VRSMI of mixed vegetation-soil pixels, FVC represents the contribution of vegetation to the cropland spectra. Based on the LSMA theory,

$$VRSMIs = \frac{VRSMI_0 - VRSMIv \times FVC}{1 - FVC}$$
$$FVC = \frac{NDVI_0 - NDVI_s}{NDVI_v - NDVI_s} \quad (4)$$

where $NDVI_0$ represents the NDVI of mixed vegetation-soil pixels, and $NDVI_v$ and $NDVI_s$ represent the vegetation NDVI and soil NDVI, respectively. Remote sensing SIs are affected by many parameters, such as vegetation leaf pigments, leaf structure, canopy structure, spatial distribution, and observation conditions. Despite this, using NDVI (or other SIs) and linearity assumptions for FVC estimation is widely accepted. Thus, Eq (4) is a simple method to remove the crop contribution to cropland VRSMI. Eq (5) shows the VRSMIs:

$$VRSMIs = \left(\frac{SWIR1 - SWIR2}{NIR \times (SWIR1)^{0.5}} - k \times FVC\right) \frac{1}{1 - FVC} \quad (5)$$

where NIR, SWIR1, and SWIR2 represent the NIR (0.83 μm), short-ware infrared bands at 1.6 μm and 2.2 μm; $k = 0.56$ is the vegetation VRSMI (Fig. 2c).

3. Datasets and methods

3.1. Lab-1 dataset: Laboratory-based leaf and soil spectra

Laboratory-based leaf spectra: Lopex1993 (by Hosgood, et al., 1993, https://teledetection.ipgp.jussieu.fr/opticleaf/lopex.htm) and angers2003 are two leaf optical property databases that are widely used for remote sensing leaf water content estimation studies. The two databases contain leaf spectra with physical and chemical parameters, such as equivalent water thickness (EWT), leaf chlorophyll *a*–b, and leaf mass per area. We used lopex1993 and angers2003 to analyze the response of the proposed VRSMI and selected the current SMIs to VWC, FVC, and SMC.

Laboratory-based soil spectra: Cinnamon, brown, and alluvial soil samples were collected from the field. Soil samples were first air-dried to a constant mass and then crushed to pass through a 2-mm screen mesh. Four steps were taken:

(a) Soil sample was placed into 12 round boxes (diameter = approximately 5 cm); the distance between the soil surface and box mouth was approximately 1 cm. The soil weight was then measured.

(b) Six water treatments (1, 2, 3, 4, 6, and 8 g) were applied to 12 soil sample boxes with two repetitions. Subsequently, all soil boxes were placed in a dark room for 30 min, and the unabsorbed liquid water on the soil surface was removed using absorbent paper.

(c) Soil weight was measured, and the soil spectra were measured using an ASD FieldSpec 3 spectrometer. The soil boxes were placed in a dark room. Repeat step 3 four times after 12 h.

(d) All soil samples were weighed after the spectral measurements. Gravimetric SMC can be calculated using the following equation:

$$SMC_g = \frac{M_0 - M_{dry}}{M_{dry}} \quad (6)$$

where SMC_g is the gravimetric SMC; M_0 and M_{dry} are the masses of the wet and air-dried soil samples, respectively; and the relative soil moisture content is

$$SMC_r = \frac{SMC_g - SMC_{dry}}{SMC_{sat} - SMC_{dry}} \quad (7)$$

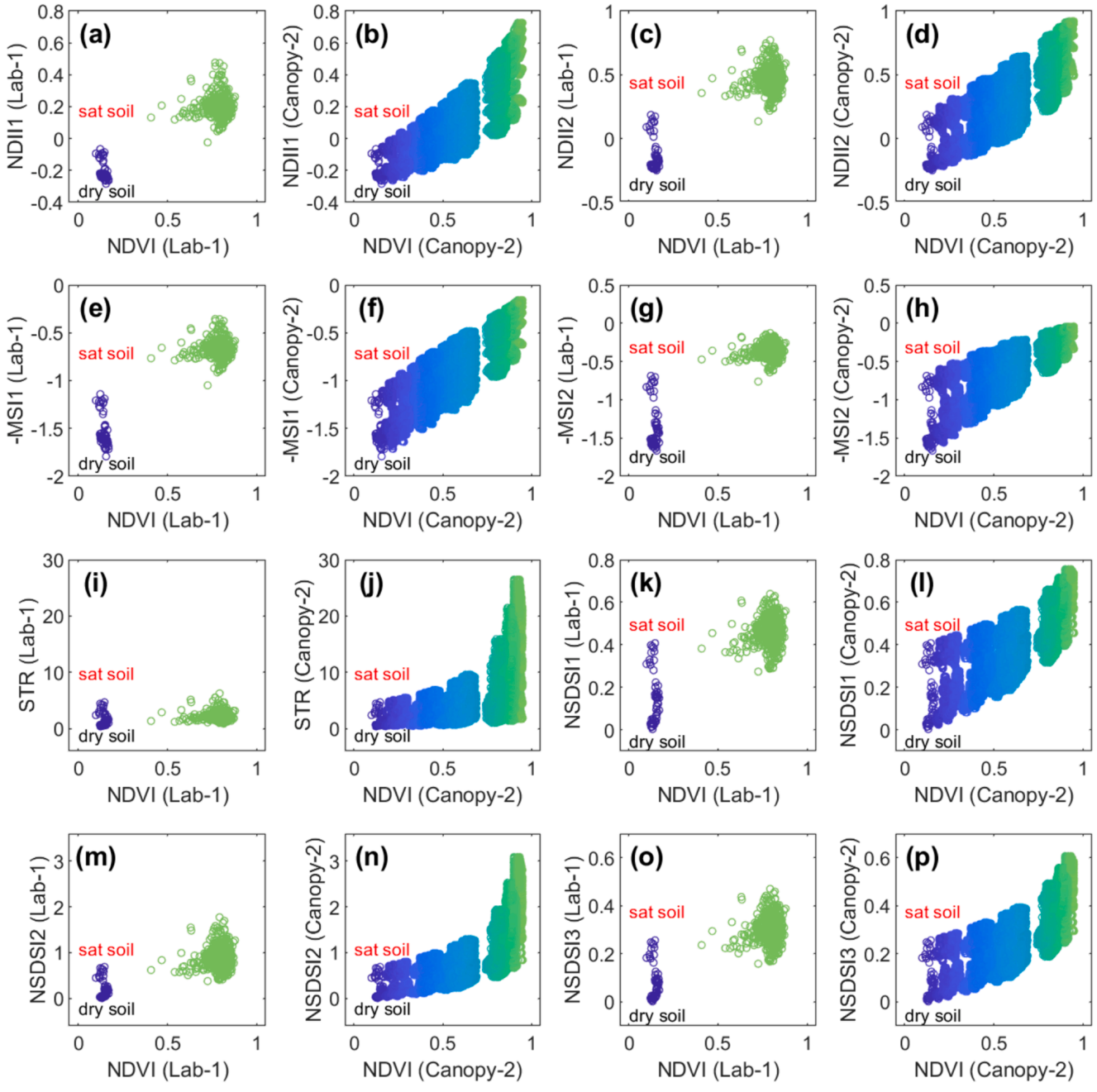


Fig. 5. Response of current SMIs to SMC, FVC, and VWC using Lab-1 and Canopy-2 datasets (cinnamon soil). (a)-(b) NDII1, (c)-(d) NDII2, (e)–(f) −MSI1, (g)-(h) −MSI2, (i)-(j) STR, (k)-(l) NSDSI1, (m)-(n) NSDSI2, (o)-(p) NSDSI3. Note: Current SMIs are increase with SMC and are very sensitive to VWC. The response of current SMIs to SMC, FVC, and VWC using brown soil (and alluvial soil) and brown soil (and alluvial soil)-based Canopy-2 dataset is similar to cinnamon soil (see Figures S1 and S2 in Supplementary materials.docx).

where SMC_r is the relative SMC content, $0 < SMC_r < 1$, and SMC_{sat} and SMC_{sat} are the gravimetric SMC of saturated and dry soil samples. After all the measurements, 60 spectra with SMC data for each soil were collected. One brown soil spectrum and one alluvial soil spectrum were identified as the outliers.

3.2. Canopy-2 dataset: Simulated vegetation canopy spectra with SMC from 0 to 1

PROSAIL 5B RTM was used to analyze the response of SMIs to SMC, FVC, and VWC. Soil spectra (from dry to saturated), leaf area index, and equivalent water thickness were set (Table 1). Parameters Cm and Cw were set based on the parameters of the lopex1993 and angers2003 databases. The leaf structure index, carotenoid content, brown pigments, hot spot, average leaf inclination angle, and observation parameters were set to fixed values. Table 1 shows the parameters set for the PROSAIL model in this study. The model parameters of PROSAIL were set based on four principles:

(1) Leaf parameters: Including leaf pigments, leaf water content, and dry matter content. These parameters are defined within a broad range (Table 1) with a small step size.

(2) Canopy and observation parameters: LAI is closely related to FVC, thus is defined within an extensive range and with a small step size, while the leaf inclination angle has a more comprehensive range and step size. Other parameters settings are based on default values from the PROSAIL RTM.

(3) Soil Background: Set to real soil spectra. A total of 178 spectra with SMC data for each soil were collected. Other numerical settings are

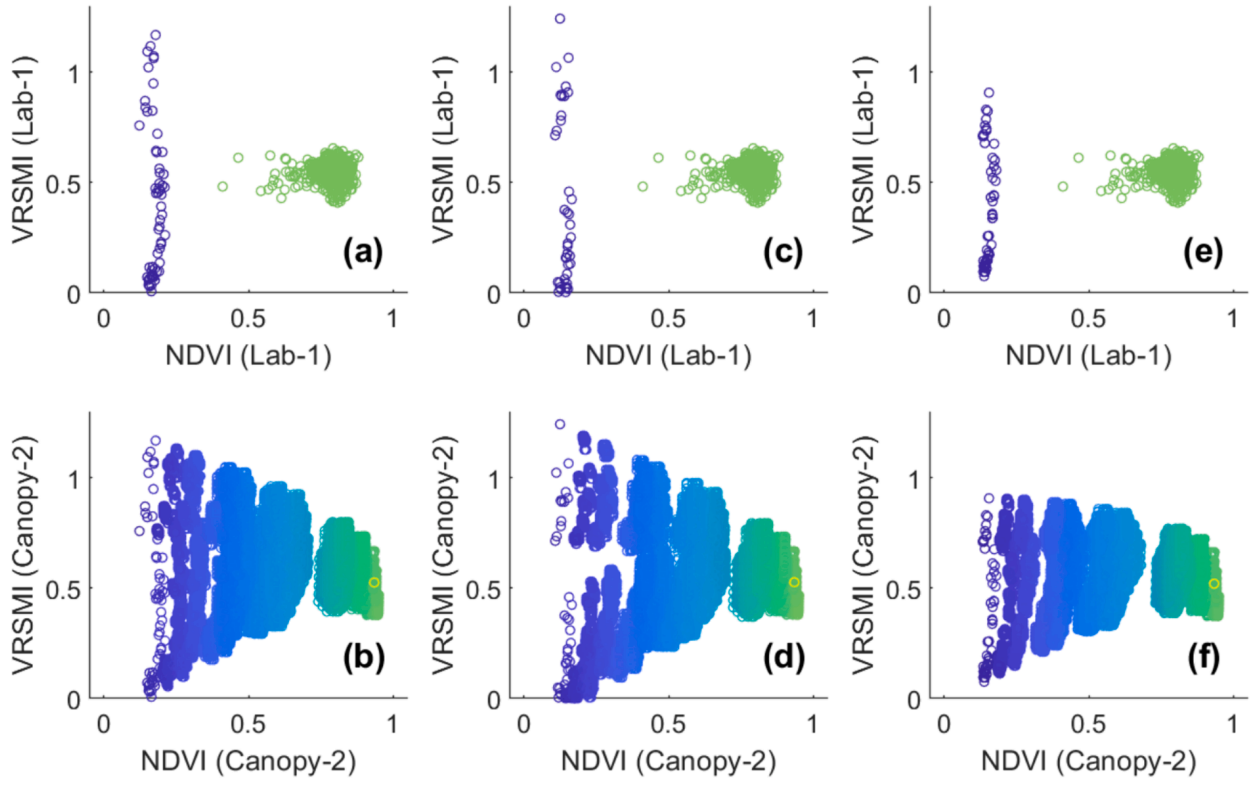


Fig. 6. Response of VRSMI to SMC, FVC, and VWC using Lab-1 and Canopy-2. (a)–(b) cinnamon soil, (c)–(d) brown soil, and (e)–(f) alluvial soil. Note: VRSMI increases with SMC and is almost insensitive (VRSMIv = 0.56 represents the average VRSMI value across all plant leaves in lopex1993 and angers2003) to VWC.

Table 5
RVS and Cor of the SMIs (based on cinnamon soil).

Index		VRSMI	MSI2	MSI1	NSDSI1	STR(SWIR2)
Lab-1	RVS (%)*	18.97	46.60	76.82	83.76	88.24
	Cor	0.24	0.89	0.90	0.77	0.20
Canopy-2	RVS (%)	30.29	65.61	102.70	108.45	302.39
	Cor	0.28	0.89	0.90	0.86	0.59
Index		**NDII2**		**NSDSI3**	**NDII1**	**NSDSI2**
Lab-1	RVS (%)*	111.36		113.65	177.95	187.99
	Cor	0.85		0.73	0.86	0.65
Canopy-2	RVS (%)	145.97		153.31	225.54	301.77
	Cor	0.89		0.85	0.88	0.80

Note: * SMIs are ranked by RVS (%) of Lab-1. The top-3 current SMIs were MSI2, NSDSI1, and MSI1. A higher RVS indicates that the SMI is more sensitive to VWC, while a lower RVS indicates that the SMI is more sensitive to SMC.

based on recommended values from the literature (Berger et al., 2018).

3.3. Field-3 dataset: Field "wheat-soil" spectra and SMC measurement

3.3.1. Study area

The study area is located at the Yuanyang County of Xinxiang City, Henan, China. Fig. 3 shows the location of the study area. Yuanyang County has a warm temperate and semi-arid monsoon climate. The average annual precipitation in the study area was approximately 490 mm. Winter wheat (October to June of the following year) and summer maize (June to September) rotation systems dominated the study area.

3.3.2. SMC and canopy spectra measurements

Field "wheat-soil" hyperspectral reflectance and soil moisture measurements were conducted from March 2 to March 14, 2024. Ground surveys were carried out in wheat fields before and after irrigation, resulting in field "wheat-soil" hyperspectral reflectance and soil moisture measurements, including dry and saturated soil samples. Fig. 3d shows the field "wheat-soil" spectra and SMC collection scenes. During the field spectra measurement, the measurement person should face the sun to ensure that no shadows are generated in the measurement area. At the same time, the spectrometer sensor was placed 1 m above the ground, and the measurement area was circular with a radius of approximately 22 cm. Before the measurement, a BaSO4 whiteboard was used to calibrate the spectrometer, and then the "wheat-soil" hyperspectral reflectance was measured. After the measurement, the BaSO4 whiteboard was re-measured. If the reflectance exceeded the range of 0.95–1.05, the measurement was repeated.

Soil moisture content was measured using the gravimetric method. Soil samples were collected directly below the location of the "wheat-soil" hyperspectral reflectance measurement. A flower shovel collected approximately 1 cm thick soil samples from the surface. The samples were placed in a sealed plastic bag. The soil samples were returned to the laboratory and weighed before being transferred to a paper bag. The paper bags were placed in an oven and dried at 90 degrees Celsius for 48 h. The samples were weighed every 12 h. The drying process was stopped when the change in soil mass between two consecutive measurements was less than 0.5 g. Then, the relative SMC can be calculated using Eq (6) and Eq (7). Table 2 shows the statistical analysis of measured soil relative SMC. Our analysis showed that the soil moisture in the study area had a large variation, with the minimum SMC value being 0 and the maximum value being 1. A total of 200 samples were obtained.

3.3.3. Spectral processing and spectral response function

A mean filter (kernel size:7) was applied to the hyperspectral reflectance measurements of the Lab-1 dataset.

$$R_i^a = \text{mean}(R_{i-3}^m + R_{i-2}^m + R_{i-1}^m + R_i^m + R_{i+1}^m + R_{i+2}^m + R_{i+3}^m) \quad (8)$$

where R_{i-3}^m, R_{i-2}^m, R_{i-1}^m, R_i^m, R_{i+1}^m, R_{i+2}^m, and R_{i+3}^m represent the hyperspectral reflectance measurements of bands i-3, i-2, i-1, i, i+1, i+2, and i+3, respectively; R_i^a is the filtered hyperspectral reflectance

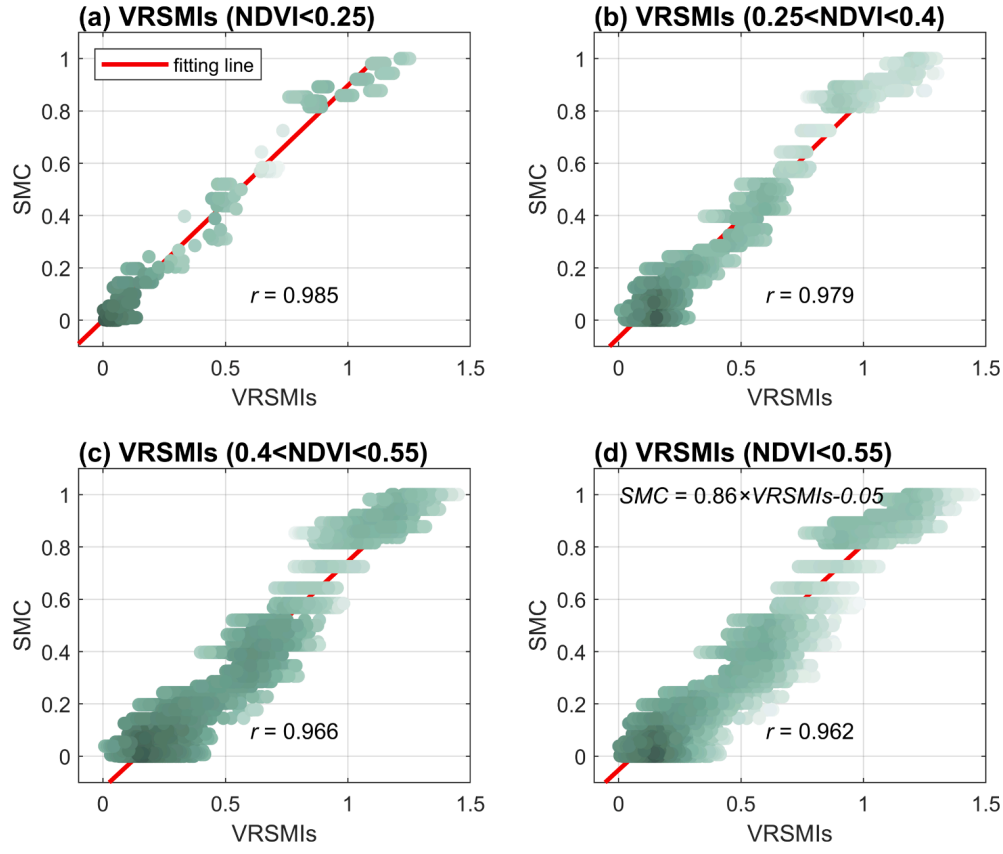


Fig. 7. Response of VRSMIs to cinnamon soil SMC under different NDVI. (a) bare soil: NDVI < 0.25, (b) low FVC: 0.25 < NDVI < 0.4, (c) moderate FVC: 0.4 < NDVI < 0.55, and (d) low-moderate: NDVI < 0.55. Note: The response of VRSMIs to SMC using brown soil (and alluvial soil)-based Canopy-2 dataset is similar to cinnamon soil (see Figures S3 and S4 in Supplementary materials.docx).

measurements. We then used a Sentinel-2 MSI spectral response function to transform the hyperspectral reflectance measurements of the Lab-1 and Canopy-2 datasets into broadband reflectances.

3.4. Current broadband soil moisture spectral indices

Many broadband remote sensing moisture indices have been proposed to estimate SMC and VWC (e.g., NDII1 (Hardisky et al., 1983), NDII2 (Hardisky et al., 1983), MSI1 (Hunt and Rock, 1989), MSI2 (Hunt and Rock, 1989), STR(SWIR2) (Sadeghi et al., 2017), NSDSI1, NSDSI2, and NSDSI3 (Yue et al., 2019) (Table 3). As the SWIR spectra are sensitive to water content, most SMIs were developed based on the SWIR bands (Table 3).

3.5. SMC model calibration and validation

3.5.1. RVS and Cor

We analyzed the SMC estimation performance of various SMI by using the SMI range ratio of vegetation to soil (RVS) and the absolute value of the correlation coefficient between the SMI and NDVI (Cor). A smaller RVS indicates that the SMI is less sensitive to VWC (Fig. 4). A lower correlation coefficient (absolute value) indicates that SMI is less affected by FVC. The RVS is defined as follows:

$$RVS = \frac{SMI^{veg}_{high-VWC} - SMI^{veg}_{low-VWC}}{SMI^{soil}_{sat} - SMI^{soil}_{dry}} \quad (9)$$

where SMI^{soil}_{sat} and SMI^{soil}_{dry} represent the SMI of saturated and air-dry soil spectra, respectively; $SMI^{veg}_{high-VWC}$ and $SMI^{veg}_{low-VWC}$ represent the highest and lowest SMI of green vegetation spectra (Lab-1: NDVI > 0.4; Canopy-2:0.85 < NDVI < 0.9).

3.5.2. Calibration and validation

Linear regression is a method that uses linear functions to model the relationship between independent variables (features) and dependent variables. We use a linear regression model to test the performance of VRSMIs in estimating SMC. For the Lab-1 dataset (lopex1993:315, angers2003:264), only the green leaf spectra satisfying NDVI > 0.4 were used in this work to analyze the robustness of new SMC index. For the Canopy-2 dataset, only the canopy spectra satisfying NDVI < 0.55 were used in the performance evaluation. For cinnamon soil-based simulation ($n_{NDVI < 0.55} = 48, 333$), for example, 60% of the Canopy-2 dataset is used to construct the linear regression model, and the remaining 40% is used to validate the model. For the Field-3 dataset (number: 200), all samples were used to validate the model. Table 4 shows the numbers of samples for SMC model calibration and validation.

4. Results and analysis

4.1. Response of VRSMI and current SMIs to SMC, FVC, and VWC based on Lab-1 dataset

Current SMIs are increase with SMC and are very sensitive to VWC. The responses of the current eight SMIs to SMC, FVC, and VWC are shown in Fig. 5. The results indicated that the current SMIs were notably sensitive to VWC (Fig. 5). For NSDSI1 (Fig. 5k, l) as an example, (a) NSDSI1 for the dry soil and saturated soil is approximately 0 and 0.4; (b) NSDSI1 for vegetation with different Cw (0.001:0.01:0.051) are approximately 0.3–0.7.

On the contrary, VRSMI increases with SMC and is almost insensitive (about 0.56 for all lopex1993 and angers2003 plant leaves) to VWC. The responses of the VRSMI to SMC, FVC, and VWC (using Cw) are shown in Fig. 6. The proposed VRSMI was almost insensitive to VWC (Cw) (Fig. 6

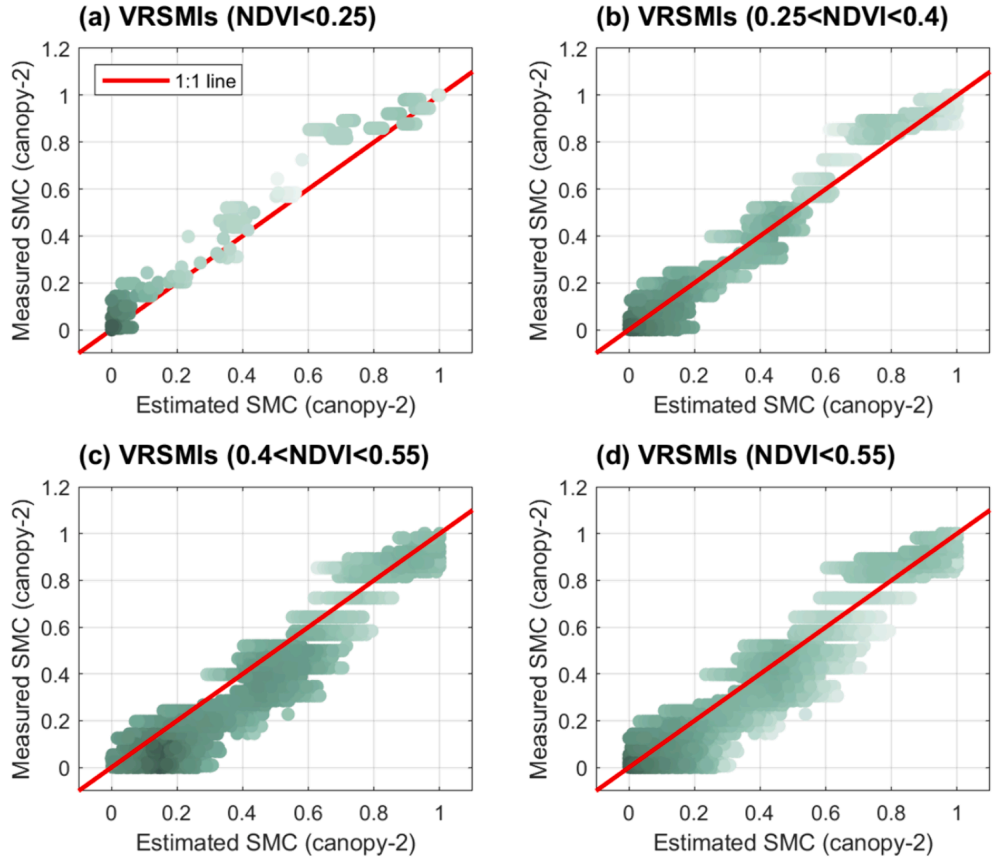


Fig. 8. Measured vs. estimated cinnamon soil SMC. (a) bare soil: NDVI < 0.25, (b) low FVC: 0.25 < NDVI < 0.4, (c) moderate FVC: 0.4 < NDVI < 0.55, and (d) low-moderate FVC: NDVI < 0.55. Note: Measured vs. estimated brown soil (and alluvial soil) SMC (see Figures S5 and S6 in Supplementary Material).

Table 6
RMSE and R^2 for four SMIs.

SMI	Scene	NDVI	R^2	RMSE	SMI	R^2	RMSE
VRSMIs*	bare soil	<0.25	**0.98**	**0.09**	MSI2	0.88	0.23
	low FVC	0.25–0.4	**0.98**	**0.07**		0.63	0.20
	moderate FVC	0.4–0.55	**0.96**	**0.11**		0.58	0.27
	low-moderate FVC	<0.55	**0.96**	**0.09**		0.46	0.24
NSDSI1	bare soil	<0.25	0.96	0.15	MSI1	0.66	0.29
	low FVC	0.25–0.4	0.90	0.10		0.30	0.26
	moderate FVC	0.4–0.55	0.87	0.18		0.27	0.30
	low-moderate FVC	<0.55	0.78	0.15		0.22	0.28

Note: R^2 and RMSE of the best SMI for the four NDVI levels are marked in bold. R^2 and RMSE of the four SMIs for brown soil (and alluvial soil) are shown in Figures S5 and S6 in the Supplementary materials. The linear relationship between (a) NSDSI1, MSI1, and MSI2 and (b) SMC are SMC = 2.45 × NSDSI1-0.15, SMC = -0.72 × MSI1 + 1.22, and SMC = -0.77 × MSI2 + 1.10.

(a)). As shown in Fig. 6, (i) VRSMI for dry soil and saturated soil are approximately 0.01 and 1.17; (ii) VRSMI for vegetation with different Cw ranges from 0.4 to 0.6. The responses of (a) VRSMI to (b) SMC, FVC, and VWC from Canopy-2 for alluvial and brown soil are similar to those of cinnamon soil and are thus not shown here (see Supplementary Materials).

Table 5 ranks the 9 SMIs by using the RVS of Canopy-2. Our results indicate that the RVS ranged from 18.97 % to 225.54 %, and the Cor ranges from 0.24 to 0.90. The top-3 current SMIs were MSI1, MSI2, and NSDSI1. The VRSMI shows a lower Cor and RVS than the top-3 current SMIs.

4.2. Validation of VRSMIs based on Canopy-2 dataset

Fig. 7 shows the response of VRSMIs to SMC (calibration dataset). The Pearson correlation coefficient (r) between VRSMIs and SMC decreased only slightly with an increase in the four NDVI levels (Fig. 7 and Table 4). The VRSMIs has great potential for detecting the SMC of croplands under low-moderate vegetation cover. The linear relationship between VRSMIs and SMC is shown in Fig. 7d (SMC = 0.86 × VRSMIs-0.05). We then estimated SMC based on linear equations (Fig. 7d). The measured vs. estimated SMC from VRSMIs is shown in Fig. 8. The estimation accuracies for the four NDVI levels were similar (Table 6; RMSE = 0.07–0.11, R^2 = 0.96–0.98). Thus, VRSMIs can help estimate SMC from low-moderate FVC regions (NDVI < 0.55).

The top-3 current SMIs (Table 5) were also selected for SMC estimation. The R^2 and RMSE for the SMC estimate from top-3 current SMIs for the four NDVI levels are shown in Table 6. The results in Table 6 indicate that: (i) the estimation accuracy of the four SMIs decreases as the NDVI increases, and (ii) the estimation accuracy of VRSMIs is noticeably higher than that of the top-3 current SMIs.

4.3. Validation of VRSMIs based on Field-3 dataset

We then estimated SMC based on linear equations (Fig. 7, SMC = 0.86 × VRSMIs-0.05, SMC = 2.45 × NSDSI1-0.15, SMC = -0.72 × MSI1 + 1.22, and SMC = -0.77 × MSI2 + 1.10). The measured vs. estimated SMC from VRSMIs and Field-3 dataset is shown in Fig. 9. The estimation accuracies for the four low-moderate vegetation coverage scenes were similar (Table 7; RMSE = 0.17–0.18, R^2 = 0.63–0.79). Thus, VRSMIs can help estimate SMC from low-moderate FVC regions (NDVI < 0.55). We also test the SMC estimation accuracies for the high FVC (using

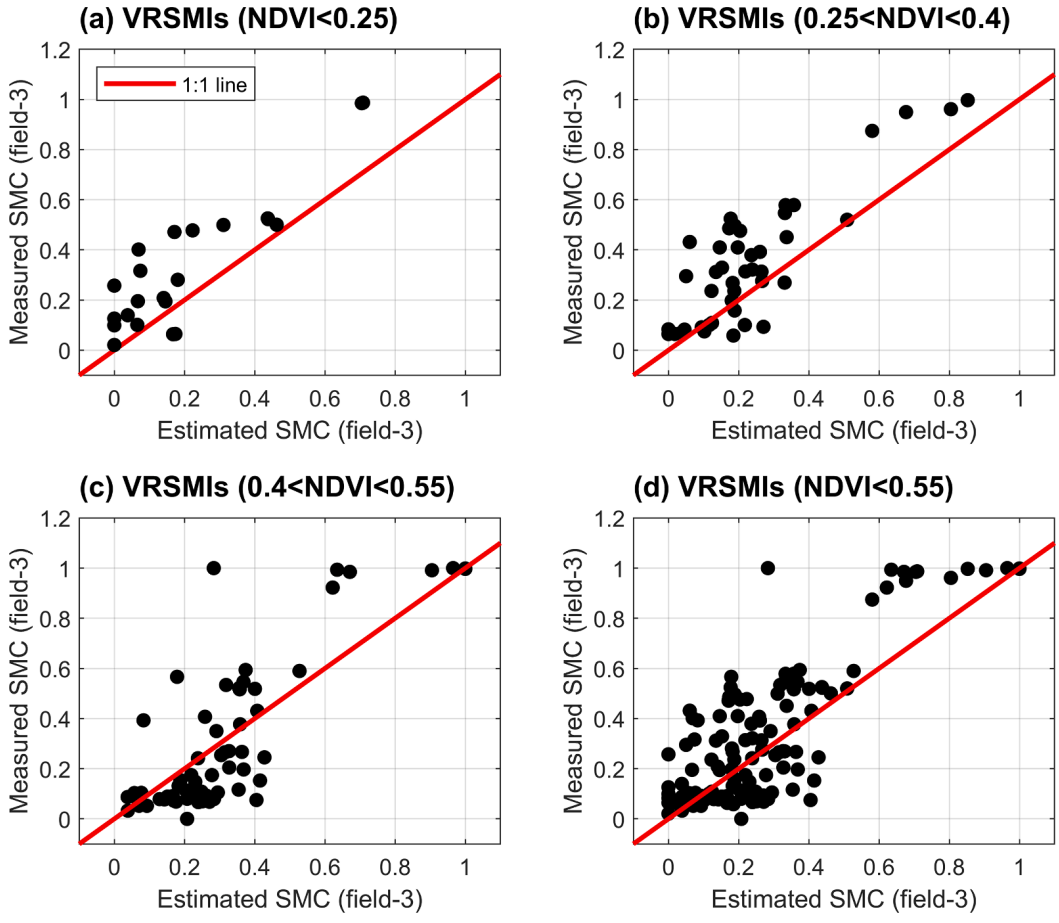


Fig. 9. Measured vs. estimated SMC. (a) bare soil: NDVI < 0.25, (b) low FVC: 0.25 < NDVI < 0.4, (c) moderate FVC: 0.4 < NDVI < 0.55, and (d) low-moderate FVC: NDVI < 0.55.

Table 7
RMSE and R^2 for four SMIs.

SMI	Scene	NDVI	R^2	RMSE	SMI	R^2	RMSE
VRSMIs*	bare soil	<0.25	**0.79**	**0.17**	MSI2	0.00	0.27
	low FVC	0.25–0.4	**0.71**	**0.17**		0.00	0.28
	moderate FVC	0.4–0.55	**0.65**	**0.18**		0.00	0.44
	low-moderate FVC	0.1–0.55	**0.63**	**0.17**		0.00	0.37
	high FVC	>=0.55	**0.36**	**0.29**		0.00	0.53
NSDSI1	bare soil	<0.25	0.63	0.27	MSI1	0.18	0.26
	low FVC	0.25–0.4	0.51	0.19		0.10	0.24
	moderate FVC	0.4–0.55	0.40	0.28		0.08	0.42
	low-moderate FVC	0.1–0.55	0.20	0.25		0.01	0.35
	high FVC	>=0.55	0.14	0.52		0.04	0.54

Note: R^2 and RMSE of the best SMI for the five NDVI levels are marked in bold and *.

NDVI>=0.55) scene. Our results suggest that the SMC estimation accuracy notably declines under high vegetation coverage scenes (Table 7; RMSE = 0.29, R^2 = 0.36).

The top-3 current SMIs (Table 7) were also selected for SMC estimation. The R^2 and RMSE for the SMC estimate from top-3 current SMIs for the four NDVI levels are shown in Table 7 (RMSE = 0.19–0.54, R^2 = 0.00–0.63). The SMC estimation accuracies of top-3 current SMIs can be ranked as NSDSI1 (RMSE = 0.25) > MSI2 (RMSE = 0.35) > MSI1 (RMSE = 0.37) based on RMSEs (low-moderate FVC scene). The results in Table 7 indicate that (i) the estimation accuracy of the SMIs decreases as the NDVI increases, and (ii) the estimation accuracy of VRSMIs is noticeably higher than that of the top-3 current SMIs.

5. Summary and conclusions

We analyzed the effects of water content on soil and vegetation reflectance. From dry to saturated, the soil spectra decreased quickly as the SMC increased (Lekner and Dorf, 2009; Philpot, 2010). For vegetation spectra, SWIR bands decrease as VWC increases (Lekner and Dorf, 2009; Philpot, 2010). Our results (Fig. 5) suggest that current optical SMIs are sensitive to cropland FVC, SMC, and crop VWC (e.g., MSI2 (Hunt and Rock, 1989), STR (SWIR2) (Sadeghi et al., 2017), NSDSI1, NSDSI2, and NSDSI3 (Yue et al., 2019)). Qualitative analysis shows that the current broadband soil moisture spectral indices (e.g., NSDSI1 in Fig. 2a and Fig. 5) are positively correlated with the NDVI and VWC, which means that vegetation water and FVC notably affect the SMC estimation based on current broadband soil moisture spectral indices.

We designed a VRSMI based on differences in the responses of the NIR and SWIR bands to water. Vegetation VRSMI was almost insensitive to VWC because vegetation SWIR1-SWIR2 and NIR×$(\text{SWIR1})^{0.5}$ decreased with VWC. Soil VRSMI is notably sensitive to SMC because soil SWIR1-SWIR2 increases with SMC when NIR×$(\text{SWIR1})^{0.5}$ decreases with SMC. As a result, the VRSMI (dry soil:0.01; saturated soil:1.17) increased with SMC and was almost insensitive to VWC (Fig. 2c, d, ranges:0.434–0.609).

Our study presents the following conclusions:

(1) The differences (Fig. 1) in the responses of NIR and SWIR bands to water help distinguish the effects of soil moisture on mixed soil-vegetation spectra. The proposed VRSMI = $\frac{(\text{SWIR1}-\text{SWIR2})}{\text{NIR}\times(\text{SWIR1})^{0.5}}$ increased with

the SMC and was almost insensitive to the VWC (Fig. 2c, d). The response of VRSMI to NDVI for lopex1993 and angers2003 datasets indicate that VRSMI is sensitive to SMC (Fig. 6, dry soil:0.01; saturated soil:1.17) and almost insensitive to crop VWC (Fig. 6, vegetation leaf with different EWT: about 0.56) in low-moderate FVC regions.

(2) VRSMIs $= \left(\frac{(SWIR1-SWIR2)}{NIR\times(SWIR1)^{0.5}} - k \times FVC\right)\frac{1}{1-FVC}$ can provide high-performance SMC estimates under low-moderate vegetation cover (NDVI $<$ 0.55). The correlation coefficient between the VRSMIs and SMC for the low and moderate -crop cover regions was similar (Fig. 7 and Table 6). We estimated SMC based on linear equations (SMC = 0.86 $\times$ VRSMIs-0.05), Canopy-2 (Fig. 8), and Field-3 dataset (Fig. 9). The SMC estimation accuracies of VRSMIs for the four NDVI levels ($<$0.25, 0.25–0.4, 0.4–0.55, $<$0.55) were similar (Table 6 and Table 7). The estimation accuracies for the four low-moderate vegetation coverage scenes were similar (Table 6; RMSE = 0.07–0.11, R^2 = 0.96–0.98; Table 7, RMSE = 0.17–0.18, R^2 = 0.63–0.79). The estimation accuracy of VRSMIs is noticeably higher than that of the top-3 current SMIs. The SMC estimation accuracies of top-3 current SMIs can be ranked as NSDSI1 (RMSE = 0.25) > MSI2 (RMSE = 0.35) > MSI1 (RMSE = 0.37) based on RMSEs (low-moderate FVC scene).

(3) The VRSMIs focus on distinguishing the vegetation-water and soil–water absorption effects on the mixed spectra within a cropland landscape (Fig. 3). Our quantitative analysis shows that the SMC estimation accuracy of VRSMIs decreases as vegetation coverage increases (Fig. 9 and Table 7). This is not surprising because (a) VRSMIs are more based on the water absorption in SWIR bands (Fig. 1), which are different from current methods like perpendicular drought index methods, (b) the contribution of soil to the mixed spectra is reduced at high vegetation cover. Optical remote sensing-based SMC estimation under high vegetation coverage has always been a problem, and this is not the focus of this study. In the future, the performance of the VRSMI index under different crop coverage scenarios should be evaluated based on more field measurements.

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CRediT authorship contribution statement

Jibo Yue: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ting Li:** Writing – review & editing, Investigation. **Yang Liu:** Writing – review & editing. **Jia Tian:** Writing – review & editing. **Qingjiu Tian:** Writing – review & editing. **Suju Li:** Funding acquisition, Writing – review & editing. **Haikuan Feng:** Writing – review & editing. **Wei Guo:** Writing – review & editing. **Hao Yang:** Writing – review & editing. **Guijun Yang:** Writing – review & editing. **Hongbo Qiao:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2024.109223.

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2024.109223.

References

Akash, M., Mohan Kumar, P., Bhaskar, P., Deepthi, P.R., Sukhdev, A., 2024. Review of estimation of soil moisture using active microwave remote sensing technique. Remote Sens. Appl. Soc. Environ. 33 https://doi.org/10.1016/j.rsase.2023.101118.

Ali, I., Greifeneder, F., Stamenkovic, J., Neumann, M., Notarnicola, C., 2015. Review of machine learning approaches for biomass and soil moisture retrievals from remote sensing data. Remote Sens. 7, 16398–16421. https://doi.org/10.3390/rs71215841.

Amani, M., Parsian, S., MirMazloumi, S.M., Aieneh, O., 2016. Two new soil moisture indices based on the NIR-red triangle space of Landsat-8 data. Int. J. Appl. Earth Obs. Geoinf. 50, 176–186. https://doi.org/10.1016/J.JAG.2016.03.018.

Amazirh, A., Merlin, O., Er-Raki, S., 2019. Including Sentinel-1 radar data to improve the disaggregation of MODIS land surface temperature data. ISPRS J. Photogramm. Remote Sens. 150, 11–26. https://doi.org/10.1016/j.isprsjprs.2019.02.004.

Ambrosone, M., Matese, A., Di Gennaro, S.F., Gioli, B., Tudoroiu, M., Genesio, L., Miglietta, F., Baronti, S., Maienza, A., Ungaro, F., Toscano, P., 2020. Retrieving soil moisture in rainfed and irrigated fields using Sentinel-2 observations and a modified OPTRAM approach. Int. J. Appl. Earth Obs. Geoinf. 89, 102113 https://doi.org/10.1016/j.jag.2020.102113.

Bablet, A., Vu, P.V.H., Jacquemoud, S., Viallefont-Robinet, F., Fabre, S., Briottet, X., Sadeghi, M., Whiting, M.L., Baret, F., Tian, J., 2018. MARMIT: A multilayer radiative transfer model of soil reflectance to estimate surface soil moisture content in the solar domain (400–2500 nm). Remote Sens. Environ. 217, 1–17. https://doi.org/10.1016/j.rse.2018.07.031.

Berger, K., Atzberger, C., Danner, M., D'Urso, G., Mauser, W., Vuolo, F., Hank, T., 2018. Evaluation of the PROSAIL model capabilities for future hyperspectral model environments: A review study. Remote Sens. 10, 85. https://doi.org/10.3390/rs10010085.

Chen, J.M., Liu, J., 2020. Evolution of evapotranspiration models using thermal and shortwave remote sensing data. Remote Sens. Environ. 237, 111594 https://doi.org/10.1016/j.rse.2019.111594.

Dall'Olmo, J., Karnieli, A., 2002. Monitoring phenological cycles of desert ecosystems using NDVI and LST data derived from NOAA-AVHRR imagery. Int. J. Remote Sens. 23, 4055–4071. https://doi.org/10.1080/01431160110115988.

ESA, 2015. SENTINEL-2 User Handbook 64. Doi: GMES-S1OP-EOPG-TN-13-0001.

Fan, L., Xing, Z., Lannoy, G.D., Frappart, F., Peng, J., Zeng, J., Li, X., Yang, K., Zhao, T., Shi, J., Ma, H., Wang, M., Liu, X., Yi, C., Ma, M., Tang, X., Wen, J., Chen, X., Wang, C., Wang, L., Wang, G., Wigneron, J.P., 2022. Evaluation of satellite and reanalysis estimates of surface and root-zone soil moisture in croplands of Jiangsu Province. China. Remote Sens. Environ. 282, 113283 https://doi.org/10.1016/J.RSE.2022.113283.

Hardisky, M.A., Klemas, V., Smart, R.M., 1983. The influence of soil salinity, growth form, and leaf moisture on the spectral radiance of Spartina alterniflora canopies. Photogramm. Eng. Remote Sens. 49, 77–83.

Hoareau, N., Umbert, M., Martínez, J., Turiel, A., Ballabrera-Poy, J., 2014. On the potential of data assimilation to generate SMOS-Level 4 maps of sea surface salinity. Remote Sens. Environ. 146, 188–200. https://doi.org/10.1016/j.rse.2013.10.005.

Hunt, E.R., Rock, B.N., 1989. Detection of changes in leaf water content using Near- and Middle-Infrared reflectances. Remote Sens. Environ. 30, 43–54. https://doi.org/10.1016/0034-4257(89)90046-1.

Ihuoma, S.O., Madramootoo, C.A., Kalacska, M., 2021. Integration of satellite imagery and in situ soil moisture data for estimating irrigation water requirements. Int. J. Appl. Earth Obs. Geoinf. 102, 102396 https://doi.org/10.1016/J.JAG.2021.102396.

Jiaxin, Q., Jie, Y., Weidong, S., Lingli, Z., Lei, S., Chaoya, D., 2024. Evaluation and improvement of temporal robustness and transfer performance of surface soil moisture estimated by machine learning regression algorithms. Comput. Electron. Agric. 217, 108518 https://doi.org/10.1016/j.compag.2023.108518.

Judge, J., Liu, P.W., Monsiváis-Huertero, A., Bongiovanni, T., Chakrabarti, S., Steele-Dunne, S.C., Preston, D., Allen, S., Bermejo, J.P., Rush, P., DeRoo, R., Colliander, A., Cosh, M., 2021. Impact of vegetation water content information on soil moisture retrievals in agricultural regions: An analysis based on the SMAPVEX16-MicroWEX dataset. Remote Sens. Environ. 265, 112623 https://doi.org/10.1016/J.RSE.2021.112623.

Karnieli, A., Agam, N., Pinker, R.T., Anderson, M., Imhoff, M.L., Gutman, G.G., Panov, N., Goldberg, A., 2010. Use of NDVI and land surface temperature for drought assessment: Merits and limitations. J. Clim. 23, 618–633. https://doi.org/10.1175/2009JCLI2900.1.

Khandan, R., Wigneron, J.P., Bonafoni, S., Biazar, A.P., Gholamnia, M., 2022. Assimilation of Satellite-Derived Soil Moisture and Brightness Temperature in Land Surface Models: A Review. Remote Sens. 14, 1–32. https://doi.org/10.3390/rs14030770.

Lei, F., Crow, W.T., Kustas, W.P., Dong, J., Yang, Y., Knipper, K.R., Anderson, M.C., Gao, F., Notarnicola, C., Greifeneder, F., McKee, L.M., Alfieri, J.G., Hain, C., Dokoozlian, N., 2020. Data assimilation of high-resolution thermal and radar remote sensing retrievals for soil moisture monitoring in a drip-irrigated vineyard. Remote Sens. Environ. 239 https://doi.org/10.1016/j.rse.2019.111622.

Lei, F., Senyurek, V., Kurum, M., Gurbuz, A.C., Boyd, D., Moorhead, R., Crow, W.T., Eroglu, O., 2022. Quasi-global machine learning-based soil moisture estimates at high spatio-temporal scales using CYGNSS and SMAP observations. Remote Sens. Environ. 276 https://doi.org/10.1016/j.rse.2022.113041.

Lekner, J., Dorf, M.C., 2009. Why some things are darker when wet. Appl. Opt. 27, 1278. https://doi.org/10.1364/ao.27.001278.

Leone, A.P., Leone, G., Leone, N., Galeone, C., Grilli, E., Orefice, N., Ancona, V., 2019. Capability of diffuse reflectance spectroscopy to predict soil water retention and related soil properties in an irrigated lowland district of southern Italy. Water (switzerland) 11. https://doi.org/10.3390/w11081712.

Li, Z., Francke, T., Zacharias, S., Schmidt, T., Schr, M., 2024. Remote Sensing of Environment Comprehensive quality assessment of satellite- and model-based soil moisture products against the COSMOS network in Germany. Remote Sens. Environ. 301, 113930 https://doi.org/10.1016/j.rse.2023.113930.

Li, J., Wang, S., 2018. Using SAR-derived vegetation descriptors in a water cloud model to improve soil moisture retrieval. Remote Sens. 10, 11–14. https://doi.org/10.3390/rs10091370.

Liang, J., Liang, G., Zhao, Y., Zhang, Y., 2021. A synergic method of Sentinel-1 and Sentinel-2 images for retrieving soil moisture content in agricultural regions. Comput. Electron. Agric. 190, 106485 https://doi.org/10.1016/j.compag.2021.106485.

Lievens, H., Tomer, S.K., Al Bitar, A., De Lannoy, G.J.M., Drusch, M., Dumedah, G., Hendricks Franssen, H.J., Kerr, Y.H., Martens, B., Pan, M., Roundy, J.K., Vereecken, H., Walker, J.P., Wood, E.F., Verhoest, N.E.C., Pauwels, V.R.N., 2015. SMOS soil moisture assimilation for improved hydrologic simulation in the Murray Darling Basin. Australia. Remote Sens. Environ. 168, 146–162. https://doi.org/10.1016/J.RSE.2015.06.025.

Liu, Y., Feng, H., Yue, J., Fan, Y., Bian, M., Ma, Y., Jin, X., Song, X., Yang, G., 2023. Estimating potato above-ground biomass by using integrated unmanned aerial system-based optical, structural, and textural canopy measurements. Comput. Electron. Agric. 213, 108229 https://doi.org/10.1016/j.compag.2023.108229.

Liu, Y., Fan, Y., Feng, H., Chen, R., Bian, M., Ma, Y., Yue, J., Yang, G., 2024a. Estimating potato above-ground biomass based on vegetation indices and texture features constructed from sensitive bands of UAV hyperspectral imagery. Comput. Electron. Agric. 220, 108918 https://doi.org/10.1016/j.compag.2024.108918.

Liu, Y., Feng, H., Fan, Y., Yue, J., Chen, R., Ma, Y., Bian, M., Yang, G., 2024b. Improving potato above ground biomass estimation combining hyperspectral data and harmonic decomposition techniques. Comput. Electron. Agric. 218, 108699 https://doi.org/10.1016/j.compag.2024.108699.

Liu, Y., Feng, H., Yue, J., Jin, X., Fan, Y., Chen, R., Bian, M., Ma, Y., Li, J., Xu, B., Yang, G., 2024c. Improving potato AGB estimation to mitigate phenological stage impacts through depth features from hyperspectral data. Comput. Electron. Agric. 219, 108808 https://doi.org/10.1016/j.compag.2024.108808.

Liu, Y., Yang, Y., 2022. Advances in the Quality of Global Soil Moisture Products: A Review. Remote Sens. 14, 1–32. https://doi.org/10.3390/rs14153741.

Liu, Z., Zhan, W., Lai, J., Hong, F., Quan, J., Bechtel, B., Huang, F., Zou, Z., 2019. Balancing prediction accuracy and generalization ability: A hybrid framework for modelling the annual dynamics of satellite-derived land surface temperatures. ISPRS J. Photogramm. Remote Sens. 151, 189–206. https://doi.org/10.1016/j.isprsjprs.2019.03.013.

Lobell, D.B., Asner, G.P., 2002. Moisture effects on soil reflectance. Soil Sci. Soc. Am. J. 66, 722–727. https://doi.org/10.2136/sssaj2002.7220.

Luo, L., Li, Y., Guo, F., Huang, Z., Wang, S., Zhang, Q., Zhang, Z., Yao, Y., 2023. Research on robust inversion model of soil moisture content based on GF-1 satellite remote sensing. Comput. Electron. Agric. 213, 108272 https://doi.org/10.1016/j.compag.2023.108272.

Moradizadeh, M., Srivastava, P.K., 2021. A new model for an improved AMSR2 satellite soil moisture retrieval over agricultural areas. Comput. Electron. Agric. 186, 106205 https://doi.org/10.1016/j.compag.2021.106205.

Peng, J., Albergel, C., Balenzano, A., Brocca, L., Cartus, O., Cosh, M.H., Crow, W.T., Dabrowska-Zielinska, K., Dadson, S., Davidson, M.W.J., de Rosnay, P., Dorigo, W., Gruber, A., Hagemann, S., Hirschi, M., Kerr, Y.H., Lovergine, F., Mahecha, M.D., Marzahn, P., Mattia, F., Musial, J.P., Preuschmann, S., Reichle, R.H., Satalino, G., Silgram, M., van Bodegom, P.M., Verhoest, N.E.C., Wagner, W., Walker, J.P., Wegmüller, U., Loew, A., 2021. A roadmap for high-resolution satellite soil moisture applications – confronting product characteristics with user requirements. Remote Sens. Environ. 252, 112162 https://doi.org/10.1016/J.RSE.2020.112162.

Philpot, W.D., 2010. Spectral reflectance of wetted soils, in: Proceedings of ASD and IEEE GRS; Art, Science and Applications of Reflectance Spectroscopy Symposium, Vol. II,.

Rahimzadeh-Bajgiran, P., Berg, A.A., Champagne, C., Omasa, K., 2013. Estimation of soil moisture using optical/thermal infrared remote sensing in the Canadian Prairies. ISPRS J. Photogramm. Remote Sens. 83, 94–103. https://doi.org/10.1016/j.isprsjprs.2013.06.004.

Reichle, R.H., Liu, Q., Koster, R.D., Crow, W.T., De Lannoy, G.J.M., Kimball, J.S., Ardizzone, J.V., Bosch, D., Colliander, A., Cosh, M., Kolassa, J., Mahanama, S.P., Prueger, J., Starks, P., Walker, J.P., 2019. Version 4 of the SMAP Level-4 Soil Moisture Algorithm and Data Product. J. Adv. Model. Earth Syst. 11, 3106–3130. https://doi.org/10.1029/2019MS001729.

Rhee, J., Im, J., Carbone, G.J., 2010. Monitoring agricultural drought for arid and humid regions using multi-sensor remote sensing data. Remote Sens. Environ. 114, 2875–2887. https://doi.org/10.1016/j.rse.2010.07.005.

Sadeghi, M., Jones, S.B., Philpot, W.D., 2015. A linear physically-based model for remote sensing of soil moisture using short wave infrared bands. Remote Sens. Environ. 164, 66–76. https://doi.org/10.1016/j.rse.2015.04.007.

Sadeghi, M., Babaeian, E., Tuller, M., Jones, S.B., 2017. The optical trapezoid model: A novel approach to remote sensing of soil moisture applied to Sentinel-2 and Landsat-8 observations. Remote Sens. Environ. 198, 52–68. https://doi.org/10.1016/j.rse.2017.05.041.

Sahaar, S.A., Niemann, J.D., Elhaddad, A., 2022. Using regional characteristics to improve uncalibrated estimation of rootzone soil moisture from optical/thermal remote-sensing. Remote Sens. Environ. 273, 112982 https://doi.org/10.1016/j.rse.2022.112982.

Santoro, M., Cartus, O., Fransson, J.E.S., 2021. Integration of allometric equations in the water cloud model towards an improved retrieval of forest stem volume with L-band SAR data in Sweden. Remote Sens. Environ. 253, 112235 https://doi.org/10.1016/j.rse.2020.112235.

Suman, S., Srivastava, P.K., Pandey, D.K., Prasad, R., Mall, R.K., O'Neill, P., 2021. Comparison of soil dielectric mixing models for soil moisture retrieval using SMAP brightness temperature over croplands in India. J. Hydrol. 602, 126673 https://doi.org/10.1016/j.jhydrol.2021.126673.

Sun, D., Kafatos, M., 2007. Note on the NDVI-LST relationship and the use of temperature-related drought indices over North America. Geophys. Res. Lett. 34, 1–4. https://doi.org/10.1029/2007GL031485.

Susha Lekshmi, S.U., Singh, D.N., Shojaei Baghini, M., 2014. A critical review of soil moisture measurement. Meas. J. Int. Meas. Confed. 54, 92–105. https://doi.org/10.1016/j.measurement.2014.04.007.

Tangdamrongsub, N., Han, S.C., Yeo, I.Y., Dong, J., Steele-Dunne, S.C., Willgoose, G., Walker, J.P., 2020. Multivariate data assimilation of GRACE, SMOS, SMAP measurements for improved regional soil moisture and groundwater storage estimates. Adv. Water Resour. 135 https://doi.org/10.1016/j.advwatres.2019.103477.

Tao, S., Zhang, X., Feng, R., Qi, W., Wang, Y., Shrestha, B., 2023. Retrieving soil moisture from grape growing areas using multi-feature and stacking-based ensemble learning modeling. Comput. Electron. Agric. 204, 107537 https://doi.org/10.1016/j.compag.2022.107537.

Tian, J., Yue, J., Philpot, W.D., Dong, X., Tian, Q., 2021. Soil moisture content estimate with drying process segmentation using shortwave infrared bands. Remote Sens. Environ. 263, 112552 https://doi.org/10.1016/J.RSE.2021.112552.

van Hateren, T.C., Chini, M., Matgen, P., Pulvirenti, L., Pierdicca, N., Teuling, A.J., 2023. On the potential of Sentinel-1 for sub-field scale soil moisture monitoring. Int. J. Appl. Earth Obs. Geoinf. 120, 103342 https://doi.org/10.1016/j.jag.2023.103342.

Vo Quang, A., Delbart, N., Jaffrain, G., Pinet, C., Moiret, A., 2022. Detection of degraded forests in Guinea, West Africa, based on Sentinel-2 time series by inclusion of moisture-related spectral indices and neighbourhood effect. Remote Sens. Environ. 281, 113230 https://doi.org/10.1016/J.RSE.2022.113230.

Wang, L., Fang, S., Pei, Z., Wu, D., Zhu, Y., Zhuo, W., 2022. Developing machine learning models with multisource inputs for improved land surface soil moisture in China. Comput. Electron. Agric. 192, 106623 https://doi.org/10.1016/j.compag.2021.106623.

Wigneron, J.P., Jackson, T.J., O'Neill, P., De Lannoy, G., de Rosnay, P., Walker, J.P., Ferrazzoli, P., Mironov, V., Bircher, S., Grant, J.P., Kurum, M., Schwank, M., Munoz-Sabater, J., Das, N., Royer, A., Al-Yaari, A., Al Bitar, A., Fernandez-Moran, R., Lawrence, H., Mialon, A., Parrens, M., Richaume, P., Delwart, S., Kerr, Y., 2017. Modelling the passive microwave signature from land surfaces: A review of recent results and application to the L-band SMOS & SMAP soil moisture retrieval algorithms. Remote Sens. Environ. 192, 238–262. https://doi.org/10.1016/J.RSE.2017.01.024.

Xu, L., Abbaszadeh, P., Moradkhani, H., Chen, N., Zhang, X., 2020. Continental drought monitoring using satellite soil moisture, data assimilation and an integrated drought index. Remote Sens. Environ. 250, 112028 https://doi.org/10.1016/j.rse.2020.112028.

Ya, Z., Binyao, W., Weiwei, Z.H.U., Li, F., Qisheng, H.E., Xin, Z., 2024. Spatial-temporal constraints for surface soil moisture mapping using Sentinel-1 and Sentinel-2 data over agricultural regions. Comput. Electron. Agric. 219, 108835 https://doi.org/10.1016/j.compag.2024.108835.

Yadav, V.P., Prasad, R., Bala, R., Vishwakarma, A.K., 2020. An improved inversion algorithm for spatio-temporal retrieval of soil moisture through modified water cloud model using C- band Sentinel-1A SAR data. Comput. Electron. Agric. 173, 105447 https://doi.org/10.1016/j.compag.2020.105447.

Yang, S., Li, S., Zhang, B., Yu, R., Li, C., Hu, J., Liu, S., Cheng, E., Lou, Z., Peng, D., 2023. Accurate estimation of fractional vegetation cover for winter wheat by integrated unmanned aerial systems and satellite images. Front. Plant Sci. 14, 1–16. https://doi.org/10.3389/fpls.2023.1220137.

Yue, J., Tian, J., Tian, Q., Xu, K., Xu, N., 2019. Development of soil moisture indices from differences in water absorption between shortwave-infrared bands. ISPRS J. Photogramm. Remote Sens. 154, 216–230. https://doi.org/10.1016/j.isprsjprs.2019.06.012.

Zappa, L., Schlaffer, S., Brocca, L., Vreugdenhil, M., Nendel, C., Dorigo, W., 2022. How accurately can we retrieve irrigation timing and water amounts from (satellite) soil moisture? Int. J. Appl. Earth Obs. Geoinf. 113, 102979 https://doi.org/10.1016/J.JAG.2022.102979.

Zhang, Y., Liang, S., Zhu, Z., Ma, H., He, T., 2022. Soil moisture content retrieval from Landsat 8 data using ensemble learning. ISPRS J. Photogramm. Remote Sens. 185, 32–47. https://doi.org/10.1016/J.ISPRSJPRS.2022.01.005.

Zhu, L., Wang, H., Zhao, T., Li, W., Li, Y., Tong, C., Deng, X., Yue, H., Wang, K., 2023a. Disaggregation of remote sensing and model-based data for 1 km daily seamless soil moisture. Int. J. Appl. Earth Obs. Geoinf. 125, 103572 https://doi.org/10.1016/j.jag.2023.103572.

Zhu, L., Dai, J., Liu, Y., Yuan, S., Qin, T., Walker, J.P., 2024. A cross-resolution transfer learning approach for soil moisture retrieval from Sentinel-1 using limited training samples. Remote Sens. Environ. 301, 113944 https://doi.org/10.1016/j.rse.2023.113944.

Zhu, W., Fan, L., Jia, S., 2023. Integration of microwave satellite soil moisture products in the contextual surface temperature-vegetation index models for spatially continuous evapotranspiration estimation. ISPRS J. Photogramm. Remote Sens. 203, 211–229. https://doi.org/10.1016/j.isprsjprs.2023.08.004.